

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Science in Information Technology (B.Sc. IT)

Semester-III University Examination November-2018

CA502 Fundamentals of Computer Networks

Date: 21.11.18, Wednesday Time: 10.00 a.m. to 01.00 p.m. Maximum Marks: 70

Instructions:

1. Attempt both sections in separate answer books.
2. The figure to the right indicates marks.
3. Make a suitable assumption when necessary.

SECTION-1

Q.1. Write the answers in brief.

[11]

1. TDM stands for _____
2. The size of logical address is _____ bytes.
3. Internet word came from _____ and _____ words.
4. Parallel communication is less costly. (True/False)
5. In OSI, _____ layer is responsible for translation, encryption, and compression of the data.
6. ISO stands for _____
7. List out 3 components of a data communication system.
8. In OSI, _____ layer is responsible for hop-to-hop delivery.
9. Optical fiber is an example of _____ medium. (Guided/Unguided)
10. In packet switching, each packet routed independently. (True/False)
11. Data link layer organizes the bits into _____. (frames/packets)

Q. 2 Answer the following questions.

[12]

- [A] Explain OSI reference model in detail (with figure).
[B] What is network? Explain its topology in detail.

[OR]

Q. 2 Answer the following questions.

[12]

- [A] List out 3 Switching Techniques. Explain any one in detail
[B] Explain 3 network criteria briefly.

Q. 3 Attempt the following questions (Any three).

[12]

- [A] Differentiate Guided Media and Unguided Media.
[B] Explain three elements of protocol briefly.
[C] Explain three modes of communication with examples.
[D] Differentiate Serial communication and Parallel Communication.

SECTION-2

Q.4. Answer the following questions.

[11]

1. Application gateway works at _____ layer in OSI reference model.
2. Error Correction is desirable on un-reliable medium. (True/False)
3. How hub is different from repeaters?

4. Open loop Congestion control mechanisms is for congestion prevention. (True/False).
5. _____ protocol is for file transfer service. (HTTP/FTP/SMTP)
6. Unacknowledged Connectionless service is appropriate for voice communication (True/False).
7. What is hamming distance?
8. In OSI, network layer transmits the data in the form of _____. (frames/packets)
9. Hub works at _____ layer.
10. Which are 2 kinds of gateways?
11. UDP is at _____ layer.

Q. 5 Answer the following questions.

[12]

- [A] Explain Odd parity bit for error detection with suitable example.
[B] Write short note on Router.

[OR]

Q. 5 Answer the following questions.

[12]

- [A] Which are two congestion control policies? Explain any one in detail.
[B] Find out CRC $[T(x)]$ for $M(x) = 11001100$ and $G(x) = 1101$.

Q. 6 Answer the following questions (Any Two).

[12]

- [A] Explain 3 categories (LAN, WAN, and MAN) of network.
[B] Define congestion. Which are various five causes of congestion in a subnet?
[C] What is framing? Explain Byte stuffing method for framing in brief with example.
[D] Explain Leaky Bucket Algorithm in detail.